



Smart & Resilient National Flag System



Resilient Tricolour Project | India Innovates Initiative

The Problem

₹104 Cr

Annual Cost

4 → 8x

Proposed Replacement Doubling

500 Flags

Across Delhi



Storm Damage

Severe summer storms tear flag fabric, requiring emergency replacements



Pollution

Delhi's AQI accelerates fabric degradation and discoloration



Manual Inspection

No real-time monitoring — damage detected only during physical visits

Dhvani: 3-Pillar Smart Flag System



Material Innovation

Kevlar-Polyester Blend + Nano Coating

4x stronger fabric, self-cleaning, 12-month lifespan vs 3 months

3x Lifespan



IoT + AI Monitoring

5 Sensors Per Mast + AI Computer Vision

Real-time tear detection, auto-alerts in <3 seconds, 94% AI confidence

Real-time



Sustainable Lifecycle

Flag Code Compliant Disposal + Recycling

100% respectful disposal with FCC certification, ₹30L/yr recycling revenue

Compliant

Material Innovation: 3x Stronger, Self-Cleaning

CURRENT VS PROPOSED

Specification	Current	Proposed
Fabric	Standard Polyester	Kevlar-Polyester Blend
Lifespan	3 months	12 months ✓
Tensile Strength	450 N/m ²	1200 N/m ² ✓
Cost per m ²	₹350/m ²	₹1,450/m ² ✓

KEY TECHNOLOGIES



Kevlar Fiber Reinforcement

15% Kevlar blend withstands 150 km/h winds



TiO2 Nano Coating

UV-activated self-cleaning reduces washing by 70%



Graphene (Future)

Phase 4

18-month lifespan, ultra-lightweight — ready for Phase 4

Smart Monitoring: Detect Damage in <3 Seconds

5 SENSORS PER MAST



Accelerometer

Vibration patterns



Strain Gauge

Fabric tension



Wind Speed

Auto-retract >80 km/h



Humidity

Moisture absorption



Camera AI-CV

YOLOv8 tear detection

4-STEP AI PIPELINE

1

CAPTURE

360° camera every 30 seconds

2

DETECT

YOLOv8 on Edge TPU, **94% mAP**

3

CLASSIFY

Minor / Major / Critical severity

4

ALERT

Push to Command Center + auto work order

COMMUNICATION STACK



LoRaWAN 868MHz



4G LTE



Edge TPU AI



<3s Alert

The Result: 86.5% Cost Reduction

~~₹104 Cr~~

Current Cost

₹43 Cr

With Dhvani

86.5%

Savings

8.5 mo

Payback Period



Investment Breakdown

₹43 Cr Year 1



Kevlar Fabric

₹31 Cr



IoT Sensors

₹6.5 Cr



AI/CV Platform

₹3.5 Cr



Software + Training

₹2 Cr



5-Year Impact

5-Year Cumulative Savings

₹305 Cr

ROI

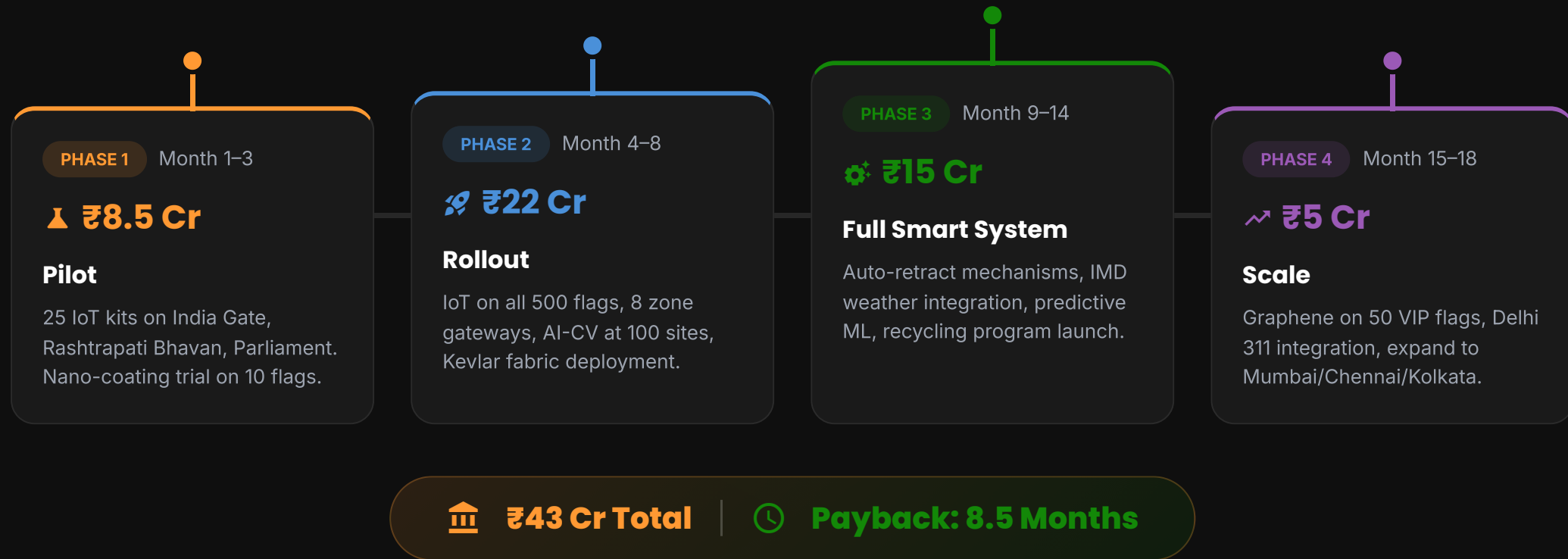
7.1x

Annual Savings

₹61 Cr/yr

18-Month Implementation Plan

Four phased deployment across Delhi's 500 high-mast flags



Dhvavini

Protecting Our Flag. Optimizing Our Resources.

 **3x Lifespan**

 **86.5% Cost Reduction**

 **<3s Alert Speed**

Team Hamaraneta | India Innovates Initiative

JAI HIND